

Po-Yun Cheng

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Research Interests: Human-Computer Interaction, Software Engineering, and AI. My broad goal is to reduce the human effort and computational cost of software development and maintenance. Specifically, I am currently interested in whether AI agents and humans need different code representations or development processes to collaborate efficiently, and whether we can systematically bridge them.

Education

Sep. 2022– Jun. 2026 **B.S. in Computer Science and Information Engineering** (GPA 3.5/4.0),
National Taiwan University, Taipei, Taiwan

- Exchange student, *University of California, Merced* (Fall 2025).
- Relevant courses: Data Structure and Algorithm, Algorithm Design and Analysis, Computer Network, Machine Learning, Software Engineering*, Deep Learning*, Advanced Human-Computer Interaction*. (*graduate level)

Research Experience

Feb. 2025– Jul. 2026 **Undergraduate Research Assistant**, Advisor: [Prof. Jonathan Lee](#)
Software Engineering Lab, National Taiwan University (remote)

- Led a three-person team on automated code review; presented a [poster](#) at Taiwan's flagship software engineering conference (TCSE 2026), winning the Best Poster Paper Award.
- Designed a stage-based multi-agent code review framework combining static analysis and project-context RAG; evaluated against CodeRabbit, GitHub Copilot, and a single-LLM ablation on real open-source project PRs, achieving the highest F1 on review-comment similarity at 90% lower inference cost (\$0.02 vs. \$0.50 per review).
- Diagnosed a vocabulary-mismatch failure in embedding-based Java retrieval and fixed it via AST-derived summary enrichment, turning zero relevant retrievals into consistent correct ones.

Apr. 2024– Oct. 2024 **Undergraduate Research Assistant**, Advisor: [Prof. Mike Y. Chen](#)
Human-Computer Interaction Lab, National Taiwan University

- Designed the formative study and questionnaire (n=16) comparing VR headset use standing vs. lying down; found supine posture reduced head rotation range by 50% horizontally and 45% vertically, defining the actuation design and target ranges.
- Built the Python analysis pipeline, applying Wilcoxon signed-rank tests and Generalized Linear Mixed Models to repeated-measures data.
- Analyzed Likert-scale outcomes (comfort, preference, dizziness) for the summative evaluation and produced all statistical figures for the resulting CHI '25 paper.

Publications

1. Wu, E.-H., **Cheng, Po-Yun**, Hsu, C.-W., Han, C. H., Lee, P. C., Fan, C.-A., Kuo, Y. C., Hu, K.-J., Chen, Y. & Chen, M. Y. *HeadTurner: Enhancing Viewing Range and Comfort of using Virtual and Mixed-Reality Headsets while Lying Down via Assisted Shoulder and Head Actuation* in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, 2025). <https://doi.org/10.1145/3706598.3714214>.

Presentations

1. **Cheng, Po-Yun**, Lee, K.-Y., Zhang, C.-W. & Lee, J. *Automated Code Review Using a Multi-LLMs Framework*
Poster presented at the 22nd Taiwan Conference on Software Engineering (TCSE 2026). **Best Poster Paper Award**. National Chin-Yi University of Technology, Taichung, Taiwan, July 2026. <https://tcse2026.seat.org.tw/tcse2026-recap>.

Work Experience

- Feb. 2026–
Present
- AI Research Intern,**
AI Solution Center, Cathay Life Insurance
- Evaluated the performance (latency, memory, accuracy) of 8 quantization settings on local LLM inference; established a selection methodology adopted across departments for model deployment decisions.
 - Built an LLM-based retrieval and clustering pipeline that surfaces insurance industry trends from 46 online sources; reducing analysts' monthly review time by 60%.

Projects

- Fall 2025
- Gesture-Based Pointer Control System and Usability Evaluation**
Advanced Human-Computer Interaction course research project [[github](#)]
- Built an end-to-end touchless pointer system that maps palm location to cursor position and thumb–finger pinch dynamics to click and drag events.
 - Designed three tasks (drag-drop, menu-select, and keyboard-input) to simulate the most frequent desktop interactions.
 - Ran a counterbalanced within-subject study (Latin square, NASA-TLX + SUS), showing mid-air gesture input underperformed a touchpad and attributing this to inter-user gesture variability and sustained-elevation fatigue.
- Fall 2025
- Graph–Temporal Convolutional Network for Real-Time Gesture Recognition**
Deep Learning course research project [[github](#)]
- Proposed GTCN, a causal graph-convolutional + dilated temporal-convolutional model for frame-wise dynamic hand gesture recognition on SHREC'21 dataset.
 - Conducted a systematic ablation over 10 architectural variants (window length, spatial/temporal pooling, classifier design), isolating future context as a driver of temporal localization (Jaccard +19%) rather than detection accuracy.

Skills

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| <i>Programming</i> | C, Java (Spring), Python (PyTorch), TypeScript/JavaScript (React) |
| <i>Tools</i> | Git, Linux, Docker, Figma, LaTeX |
| <i>Languages</i> | Mandarin (native), English (TOEFL iBT 5.0/6.0; R=5.0, L=5.0, W=5.0, S=5.0) |